C-HI 5 Myr Aligned Comparison And Determinism Gate

Date: 2026-05-10

Summary

This report folds the aligned `C-HI` 5 Myr, `0 deg` smoke products onto one page and records the deterministic particle-radio cross-check requested before using the lane for longer snapshots.

The aligned target is:

- case: `C-HI`, high-power Stewart cluster control
- grid output: `data.0008.flt.h5`
- grid time: `4.9994 Myr`
- view: `0 deg`, projected along the simulation `y` axis
- truth threshold: `tracer > 5e-3`
- radio frequency: `150 MHz`
- X-ray energy: `1 keV`

Aligned Products

The same target now has:

- Level 1 injected-energy truth
- Level 2 tracer-defined lobe truth
- ellipsoidal cavity-boundary and `pV` smoke
- XRAiSE-corrected 1 keV grid-thermal X-ray map
- particle-loss-aware 150 MHz radio/lobe map

The compact comparison assets are stored in:

```
docs/science/assets/c_hi_aligned_comparison_2026-05-10/
```

Scalar Comparison

Quantity	Value
grid time	`4.9994358508106655 Myr`
one-sided elapsed injected energy	`1.5777e59 erg`
bipolar elapsed injected energy	`3.1554e59 erg`
Level 2 tracer-lobe volume	`10791.775 kpc^3`

Quantity	Value
Level 2 tracer-lobe total energy	1.8459e59 erg
bipolar cavity pV	6.3419e58 erg
bipolar cavity 4pV	2.5367e59 erg
mean cavity sound age	13.4484 Myr
XRAiSE-corrected 1 keV map max	2.8783e6 mJy sr^-1
XRAiSE-corrected 1 keV map sum	1.0937e9 mJy sr^-1 pixels
particle-loss 150 MHz map max	3.6847e15 mJy sr^-1
bipolar lossy 150 MHz luminosity proxy	6.3450e27 W Hz^-1
positive/negative radio lobe balance	0.9851

The energy ordering is physically sensible for this early symmetric control: the tracer-defined lobe energy is below the elapsed injected energy, while the simple cavity 4pV proxy sits between the Level 2 lobe energy and the total bipolar injection.

Comparison Figure

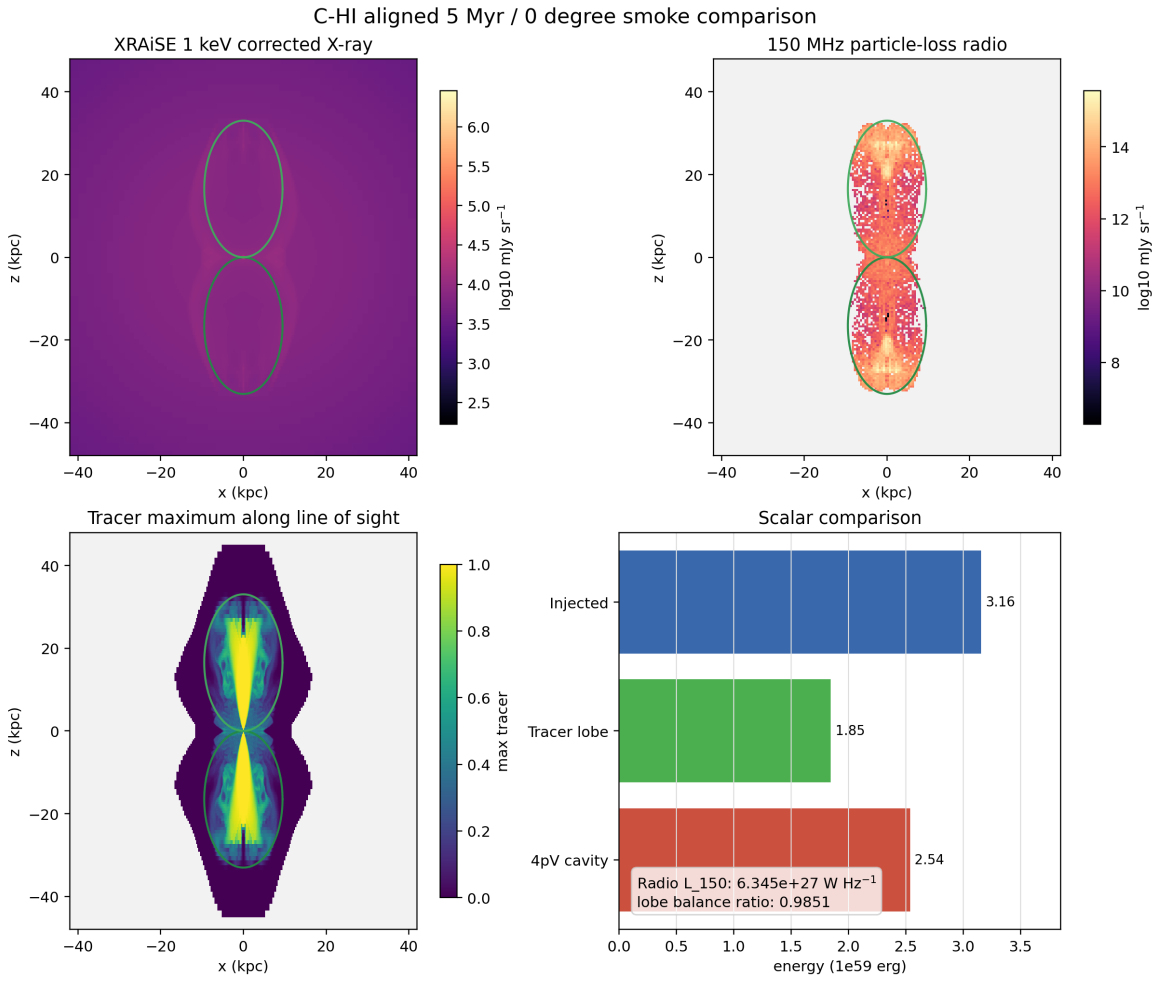


Figure 1: C-HI aligned comparison maps

The green ellipses are the first cavity-boundary stub from the same `tracer > 5e-3` target. The bottom-right panel puts the key truth and cavity energy scales on the same axis, while the text box records the radio luminosity and lobe-balance smoke.

Determinism Check

The supplied `xraise.py` particle-loss path still emits a `NumbaParallelSafetyWarning` inside `praise_aio(...)`. To test whether this warning was producing thread-dependent results, the same particle-radio smoke was rerun on `kunanyi` with:

- NUMBA_NUM_THREADS=1
- NUMBA_NUM_THREADS=4

The PBS job was:

- job id: 690644.kunanyi-ohpc
- job name: c_hi_rdet5
- node: n001
- resources: select=1:ncpus=4:mpiprocs=1
- archive: /u/bwbrook/codex_kunanyi/c_hi_particle_radio_determinism_2026-05-10.tar.gz
- local archive: build/science/kunanyi_stage_2026-05-08/c_hi_particle_radio_determinism_690644-05-10.tar.gz
- archive SHA256: e04a16b996111a118998eac52fd3c10f90a602b0fc9a54a8d9df3e28b71802f0

Result:

Check	Result
arrays allclose at rtol = 1e-10	True
exact output hashes match	True
maximum array relative difference	0.0
maximum scalar-metric relative difference	0.0
overall pass	True

This does not prove the supplied Numba warning is harmless for every future configuration, but it does show that the current 5 Myr, 150 MHz particle-radio smoke is bitwise stable between one and four Numba threads.

Determinism Figure

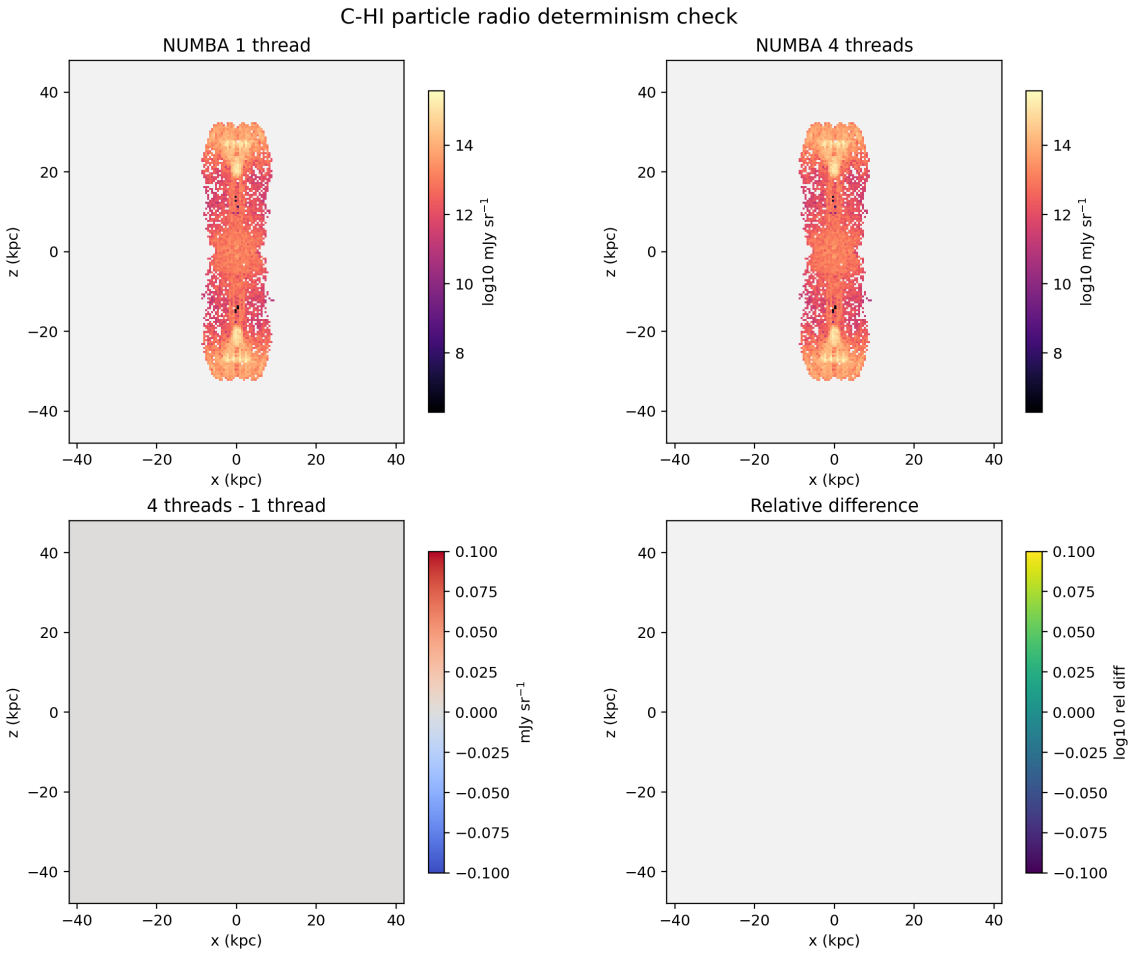


Figure 2: C-HI particle radio determinism check

The lower panels are blank because the difference is exactly zero in the saved products.

Products

Aligned comparison products:

- `docs/science/assets/c_hi_aligned_comparison_2026-05-10/aligned_smoke_metrics.csv`
- `docs/science/assets/c_hi_aligned_comparison_2026-05-10/c_hi_aligned_comparison_maps.png`

Determinism products:

- docs/science/assets/c_hi_particle_radio_determinism_2026-05-10/determinism_summary.csv
- docs/science/assets/c_hi_particle_radio_determinism_2026-05-10/determinism_summary.json
- docs/science/assets/c_hi_particle_radio_determinism_2026-05-10/determinism_array_diff.cs
- docs/science/assets/c_hi_particle_radio_determinism_2026-05-10/determinism_metric_diff.c
- docs/science/assets/c_hi_particle_radio_determinism_2026-05-10/determinism_hashes.csv
- docs/science/assets/c_hi_particle_radio_determinism_2026-05-10/c_hi_particle_radio_deter

Reusable scripts:

- scripts/science/build_aligned_smoke_comparison.py
- scripts/science/compare_particle_radio_determinism.py
- scripts/kunanyi/c_hi_particle_radio_determinism.pbs

Interpretation

This closes the first aligned-observable gate for the early C-HI control snapshot. The project now has one declared 5 Myr target where truth, cavity, X-ray, particle-loss, and radio products can be compared directly, and the particle-radio lane is stable across the tested Numba thread counts.

The next science move should be to extend the same extraction/reporting path to the next early snapshots, then decide whether the current C-HI run should be continued toward the first 50 Myr milestone or whether the grid/output settings need adjustment before spending that larger compute block.